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EMPATHY AND PERCEPTIONS OF FEMALE SPEAKERS VARYING IN CONFIDENCE

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Abstract – We manipulated the confidence of a female speaker and assessed listeners' characteristics and perceptions of the speaker. We hypothesized that the speaker would be perceived more positively and information recalled more accurately when the speaker was confident. We also hypothesized that listeners high in empathy would evaluate the low confident speaker less harshly. Participants were randomly assigned to one of two experimental conditions and watched a 3-minute video of a female speaker describing a research study. The speaker presented as either high confidence or low confidence. We assessed participants' perceptions of the speaker, recall, confidence in recall, cognitive empathy, and affective empathy. We achieved mixed support for our hypotheses. As predicted, the confident speaker was viewed more positively and less negatively; however, she did not elicit better recall. Listeners' empathy did not predict their perceptions with one exception. Listeners higher in affective empathy evaluated the low confidence speaker most negatively and the high confidence speaker least negatively. Men had more accurate recall of the presented information than did women, which likely is a result of men having more prior research experience. These findings support previous research that demonstrate the salience of speaker confidence, which may be even more influential than listener characteristics in driving perceptions.

Conveying confidence is a valuable characteristic in social contexts. An individual who exhibits confidence typically experiences higher social status, is seen as more trustworthy, has greater persuasive power, and is perceived as more knowledgeable (Roche et al., 2023). One way to convey confidence is through speech. Confident speakers are perceived as more committed and believable (Jiang et al., 2020; Mori & Pell, 2019). Listeners are more willing to trust and learn from speakers who convey confidence (Birch et al., 2020; Pozzi & Mazzarella, 2023). The "confidence heuristic" is a term used to reflect the reliably demonstrated fact that listeners' prefer and attribute more positive characteristics to confident, versus uncertain, speakers (Birch et al., 2020; Thomas & McFadyen, 1995).

Speaker confidence is conveyed to listeners through both verbal and non-verbal cues. For example, stating information in a factual manner and avoiding meaningless hand gestures convey confidence, while a slumped posture and distressed facial expression convey a lack of confidence (Pozzi & Mazarella, 2023). Tentative wording such as "I guess" are even more powerful signals than are non-verbal cues that a speaker lacks confidence (Tenney et al., 2019). Mori and Pell (2019) asked observers to guess speakers' confidence levels, while the speakers engaged in tasks that induced either high or low confidence. Even with the audio muted, observers were accurate at identifying speaker confidence. One biological theory to explain the preference for confident speakers is that the vocal characteristics associated with confidence lead to deeper cognitive processing (Guyer et al., 2021). Jiang and Pell (2015) found that confident speakers elicited stronger event-related potentials in listeners' brains than did uncertain speakers.

In general, female speakers are perceived as less confident than male speakers (Roche et al., 2019, 2022). Female speakers also carry a heavier burden than male speakers, because the manner in which they convey information is more influential in determining listener perceptions. Roche et al. (2022) demonstrated that female speakers are viewed as more uncertain than male speakers, and this difference becomes even more pronounced when female speakers use rising intonation (i.e., a questioning tone) at the end of their sentences. Similarly, male and female speakers are perceived as equally agreeable unless the female speaker pauses before an affirmative statement, in which case her agreeableness rating declines (Roberts & Norris, 2016). Men maintain their ratings of intelligence regardless of whether they speak with vocal fry (also known as creaky voice);

however, women's intelligence ratings decrease if they speak with vocal fry (Taylor et al., 2022). Women may have some intuitive understanding of this gender bias, because women are less likely to give a public presentation when provided the opportunity than are men, even when incentives are offered (DePaola et al., 2021).

Listener characteristics can also influence perceptions of speakers. Much of the research in this area focuses on the linguistic background of the listener and whether it matches, or does not match, the linguistic presentation of the speaker. For example, listeners' familiarity with non-native speech influences their perceptions of non-native speakers (Woolridge et al., 2023). Outside of this particular research framework, there is limited and inconsistent evidence about whether listener characteristics affect perceptions to the extent that speaker characteristics do. Hancock and Pool (2017) recruited cisgender men, cisgender women and transgender women, of varied sexual orientations, to evaluate auditory recordings made by cisgender men, cisgender women, and transgender women. The only significant consequence of listeners' demographics was that heterosexual listeners, compared to non-heterosexual listeners, perceived the transgender women as slightly less feminine. In other words, listeners' gender and sexual orientation were not salient variables in driving their perceptions of a speaker. More recent researchers found no effect of listener gender on attributions made about speakers' gender (Brown et al., 2021b; Hancock et al., 2023). As further evidence that individual listener characteristics might exert minimal influence, researchers have observed that listeners often display a high level of agreement when evaluating speakers. This consensus among listeners persists even when their perceptions are inaccurate, suggesting that listeners rely on shared biases or common evaluative criteria (e.g., Mahrholz et al., 2018; Welch et al., 2021).

One exception is that an extensive body of research addresses the listener characteristic of empathy. However, the focus has been on the importance of active and empathetic listening in interpersonal situation such as between doctor and patient (e.g., Junaid et al., 2022), between mental health clinician and client (e.g., Aadam et al., 2024), or in social dyads (e.g., Brown et al., 2021a). Empathy can exist as either a stable trait or elicited, and both facilitate these social interactions (Jankowiak-Siuda et al., 2011). Individuals with either type of empathy show enhanced responsiveness to speakers, particularly when the topic has an emotional component. For example, listening to vocal expressions of pain result in increased activity in areas of the brain associated with affective and cognitive aspects of empathy (Mauchand et al., 2023). In a modern twist, Carolus et al. (2021) found that observers reported greater empathy for an Amazon Echo, a voice-activated virtual assistant device, when a research confederate spoke to it rudely versus neutrally.

Despite the recognition that empathy impacts listeners' physiology and emotions, limited research has assessed whether listeners' empathy predicts perceptions of speaker characteristics. Our study brought together these two lines of research. We examined perceptions of a young adult female speaker who conveyed varying degrees of confidence while addressing a young adult audience. We also examined the influence of listeners' ability to recognize the feelings of others (cognitive empathy) as well as their ability to feel the feelings of others (affective empathy) on these perceptions. We hypothesized that:

1. compared to the low confidence (LC) condition, participants in the high confidence (HC) condition would agree more with positive attributes for the speaker and agree less with negative attributes for the speaker.
2. compared to the low confidence (LC) condition, participants in the high confidence (HC) condition would have more accurate recall of presented information.

These predictions align with the "confidence heuristic" suggesting that listeners are more inclined to trust and absorb information from speakers who exhibit confidence, as well as recent research suggesting that confident speakers elicit deeper cognitive processing in listeners. We also hypothesized that:

3. compared to listeners low in empathy, listeners high in either affective or cognitive empathy would evaluate the LC speaker less harshly.

This prediction is based on the idea that individuals with higher empathy are more attuned to and understanding of others' emotional states. This heightened empathy may lead them to be more forgiving and supportive, resulting in more consistent judgments of the speaker irrespective of confidence condition.

Method

Participants

Participants (n = 57) were currently enrolled college students recruited through an undergraduate classroom. The sample had a mean age of 18.52 (*SD* = .84). Among the participants, 32 identified as Caucasian, 17 as African American, 5 as multi-racial, and 2 as Hispanic/Latino. In terms of gender, 42 participants identified as women, 10 as men, 2 as non-binary, and 2 as unspecified "other." Regarding sexual orientation, 31 participants reported being heterosexual, 17 identified as

bisexual, 6 as homosexual, and 2 identified as unspecified "other." A single participant responded to the survey but did not provide any demographic data. This project was approved by the university's Institutional Review Board. Students were offered extra credit for participation and could select an alternate extra credit assignment if they chose not to participate. Thus, all participation was voluntary.

Materials and Procedures

To create two randomized groups, students in the original classroom were asked to count off one and two. One of the groups was then relocated to a different classroom across the hallway. The classrooms were selected because they were in the same hallway and had the same video equipment and seat configuration. The researchers confirmed that the sound levels and picture quality were equivalent in the two classrooms.

Participants were asked to make sure nothing was on their desk or in their hands. They were then directed to watch a three-minute video displayed on a large overhead screen at the front of the classroom. One of the groups viewed a peer-aged, female speaker describing the methodology and results of a research study she had conducted. She spoke confidently with no verbal crutches or hesitations, creating our High Confidence condition (HC; n = 28). See the appendix for the script. The other group of participants watched a video of the same female speaker following the same script; however, in this Low Confidence condition (LC; n = 29), the speaker inserted pauses, oral stumbles, and apologies into the script. These modifications added approximately 12 seconds to the length of the video. To ensure consistency, the two videos were recorded consecutively on the same day with the same camera, lighting, background, and outfit.

Immediately after the video ended, participants were provided with a link to an online Qualtrics survey and accessed it through their cell phones. We provided the following instructions, "We are interested in your personal opinions of the video you just watched. Please give us your honest opinion of how well the speaker matches the following descriptions." Participants encountered six positive characteristics (intelligent, confident, trustworthy, professional, impressive, interesting) and five negative characteristics (inexperienced, incompetent, unethical, hard to understand, cringy). The presentation of the characteristics was randomized such that negative and positive characteristics were not grouped together. All participants viewed the characteristics in the same randomized order. Participants were asked to rate the extent to which each one described the speaker with 1 representing "strongly disagree" and 5 representing "strongly agree." These items were inspired by the Muenster Epistemic Trustworthiness Inventory (Hendriks et al., 2015). Participants received a mean score for the positive adjectives and a mean score for the negative adjectives, with a possible range of one to five. We achieved a Cronbach's Alpha of .85 for the positive traits and .80 for the negative traits. We asked participants to make ratings of one additional descriptor (attractive) to assess whether the physical perceptions of the speaker influenced judgments of her ability.

Next, we provided an open-ended field and asked participants what the speaker did best and how the speaker could improve. Participants were then provided with the statement, "I think I could have done a better job conveying the information." To assess participants' confidence in the accuracy of their memory recall, we provided the statement, "I accurately remember what the speaker talked about in this video." They responded with their level of agreement using a 5-point scale where 1 represented "strongly disagree" and 5 represented "strongly agree."

Participants then encountered a five-item quiz to assess how much information they accurately recalled from the video. One item assessed the dependent variable in the study, while the others assessed memory of the study's hypotheses and results. Participants could earn a score from zero to five, with a higher score indicating more accurate recall of the video's content.

Participants then encountered the 20-item Perth Empathy Scale (Brett et al., 2023), which assesses the trait of empathy. This scale can be divided into two sub-scales. The cognitive empathy subscale measures the ability to recognize the feelings of others. A sample item was "Just by seeing or hearing someone, I know if they are feeling sad." The affective empathy subscale measures the tendency to feel the feelings of others. A sample item was, "When I see or hear someone who is sad, it makes me feel sad too." The published reliability for cognitive empathy is .92, and for affective empathy is .80 (Brett et al., 2023). We achieved a Cronbach's Alpha of .95 for cognitive empathy and .86 for affective empathy.

We asked participants four demographic questions, including age, gender, race/ethnicity, and sexual orientation. In order to further understand our participants, we also asked them to quantify how much experience they personally had conducting scientific research on a five-point scale where 1 represented "never" and 5 represented "extensive."

Results

We divided the sample into high and low cognitive empathy by using a median split with cognitive

empathy scores. We ran a 2 (condition) X 2 (cognitive empathy) MANOVA. There were no significant interactions in how listeners ascribed positive traits [$F(3,53) = .20$, $p = .66$, *ns*] or negative traits [$F(3,53) = .32$, $p = .57$, *ns*] to the speaker. There were also no interactions related to listeners' recall accuracy [$F(3,53) = .008$, $p = .93$, *ns*], confidence in recall accuracy [$F(3,53) = 2.40$, $p = .13$, *ns*], or belief they could have done a better job [$F(3,53) = .06$, $p = .81$, *ns*]. We found main effects for condition. Compared to listeners who saw the LC speaker, those who saw the HC speaker expressed higher agreement for positive traits [$F(3,53) = 44.21$, $p < .001$], lower agreement for negative traits [$F(3,53) = 34.22$, $p < .001$], and less agreement that they could have done a better job [$F(3,53) = 11.87$, $p = .001$]. There was no main effect for condition related to recall accuracy, confidence in recall accuracy, or attractiveness of the speaker. The comparison between the HC and LC conditions for individual trait variables and recall accuracy can be seen in Table 1. We found no main effects for cognitive empathy on any of the variables.

We then divided the sample into high and low affective empathy by using a median split with affective empathy scores. We ran a 2(condition) X 2 (affective empathy) MANOVA. We found a significant interaction for negative traits, with listeners high in affective empathy agreeing more with negative traits for the LC speaker and agreeing less for the HC speaker [$F(3,53) = 5.04$, $p = .03$]. The interactions were not significant for positive traits [$F(3,53) = 1.65$, $p = .21$, *ns*], recall accuracy [$F(3,53) = 3.86$, $p = .06$, *ns*], confidence in recall accuracy [$F(3,53) = .01$, $p = .94$, *ns*], or belief they could have done a better job [$F(3,53) = .02$, $p = .88$, *ns*]. There were no main effects for affective empathy. The data related to affective and cognitive empathy can be seen in Table 2.

Table 1

Means and Standard Deviations for Speaker Attributes and Recall Accuracy in HC and LC Conditions

	High Confidence Speaker	Low Confidence Speaker
Intelligent	4.29 (.90)*	3.83 (.89)
Confident	4.50 (.58)**	2.24 (1.30)
Trustworthy	4.04 (.69)**	3.24 (1.15)
Professional	4.29 (.60)**	2.59 (1.40)
Impressive	3.93 (.66)**	2.76 (1.06)
Interesting	3.89 (.96)*	3.14 (1.33)
Attractive	2.79 (1.10)	2.72 (.96)
Inexperienced	2.18 (1.02)	3.62 (1.27)**
Incompetent	1.68 (.82)	2.59 (1.27)**
Unethical	1.89 (1.10)	2.34 (1.23)
Hard to Understand	1.61 (.88)	3.72 (1.22)**
Cringy	2.61 (1.10)	3.34 (1.29)*
Recall Accuracy	2.04 (.88)	1.86 (.74)

* $p < .05$
** $p < .01$

Table 2

Means and Standard Deviations for Dependent Variables Across Conditions and Affective Empathy

	High Confidence Speaker		Low Confidence Speaker	
	Negative Traits	Positive Traits	Negative Traits	Positive Traits
Low Affective Empathy	2.35 (.66)	3.94 (.35)	3.19 (.78)	2.93 (.42)
High Affective Empathy	1.63 (.60)	4.40 (.38)	3.42 (1.0)	2.92 (1.11)
Low Cognitive Empathy	2.13 (.81)	4.06 (.38)	3.32 (.87)	2.89 (.51)
High Cognitive Empathy	1.88 (.61)	4.27 (.46)	3.32 (.96)	2.94 (1.05)

Because we did not find categorical differences related to accuracy or confidence in recall accuracy, we examined the data file as a whole. The more confidence participants had in the accuracy of their recall, the: fewer negative traits they ascribed to the speaker [($r(57) = -.42$, $p < .001$] and the more positive traits they ascribed [$r(57) = .33$, $p = .01$)]. The more confidence participants had in the accuracy of their recall, the more prior research experience they had [$r(56) = .31$, $p = .02$]. These represent weak to moderate correlations.

We compared participants who identified as men with those who identified as women using an independent samples *t*-test. Compared to women, men more accurately recalled information from the two videos [$t(50) = 2.22$, $p = .02$]. There was a trend for men to express more confidence in the accuracy of their recall than women, although it fell just short of significant [$t(50) = 1.57$, $p = .06$, *ns*]. Men reported more prior research experience than did women [$t(50) = 1.94$, $p = .03$]. We found no gender difference in the attractiveness rating for the speakers [$t(50) = -1.61$, $p = .06$, *ns*]; in fact, female participants rated the female speaker slightly (but not significantly) more attractive than the male participants.

We compared African American and Caucasian adults using an independent *t*-test. We found no differences in the accuracy of their memory recall [$t(47) = .82$, $p = .21$, *ns*], confidence in memory recall [$t(47) = .53$, $p = .30$, *ns*], or prior research experience [$t(47) = -1.36$, $p = .09$, *ns*]. Age did not predict any variables of interest.

Two raters categorized the qualitative comments, with an inter-rater reliability of .97. A summary of the data can be seen in Table 3. First, the raters examined the patterns for the two conditions on what the speaker did best. For the HC speaker ($n = 28$), 42.8% of compliments related to "explained research well" (e.g., communicate their data and hypothesis clearly and effectively"), and 25% related to "clear message and speaking" (e.g., "annunciate and clearly communicate points"). For the LC speaker responses ($n = 29$), 58.6% related to "explained research well" (e.g., "giving data to support her hypothesis and explaining findings"), and the next highest category was 13.7% of participants providing a criticism of the speaker (e.g., "I don't know, everything seemed terrible.").

The raters then categorized the qualitative comments related to what could be improved. For the HC speaker ($n = 28$), 25% of the suggestions related to "increased eye contact" (e.g., "she should make more eye contact") followed by 17.8% for "provide more detail" (e.g., "include percentages and number within results") and "nothing" (e.g., "no idea"). For the LC speaker responses ($n = 29$), 27.6% related to "be more confident" (e.g., "the speaker can improve by being a bit more confident in what they are talking about"), and the next highest category was 20.7% of participants pointing out one of the low confidence behaviors (e.g., "stuttering, stopping to say I'm [sic] sorry, hand gestures"). This qualitative data serves as a validity check, providing evidence that participants differentiated between the high and low confidence conditions.

Table 3
Percentage of Participants Providing Responses in Each Rater-Created Category of Qualitative Responses

	High Confidence Speaker	Low Confidence Speaker
Did Best	Explained research well (42.9%) Clear message/speech (25.5%) Confident (10.7%) Voice tone/pace (10.7%) Concise/Focused (7.1%) Eye contact (3.7%)	Explained research well (58.6%) Nothing/provided criticism (13.8%) Corrected mistakes (10.3%) Friendly face/smile (10.3%) Loud speaking tone (6.9%)
Could Improve	Increase eye contact (25%) Nothing (17.9%) Provide more detail (17.9%) Be more engaging (14.3%) Talk slower (10.7%) Relax/ be less stiff (10.7%) Be less confident (3.5%)	Be more confident (27.6%) Avoid verbal crutches (20.7%) Know information better (17.2%) Be clearer (17.2%) Relax/ calm down (13.8%) Increase eye contact (3.4%)

Discussion

We hypothesized that compared to the low confidence condition, participants in the high confidence condition would agree more with positive attributes for the speaker and agree less with negative attributes for the speaker. Our data supported these hypotheses. As predicted, the high confidence speaker received significantly higher positive ratings and lower negative ratings. The qualitative comments from participants matched this pattern. The favorable impressions of the high confidence speaker match previous research documenting that listeners attribute a wide range of positive attributes to confident, versus uncertain, speakers (e.g., Birch et al., 2020; Jiang et al., 2020; Pozzi & Mazzarella, 2023).

We also hypothesized that compared to the low confidence condition, participants in the high confidence condition would have more accurate recall of presented information. This hypothesis was not supported. The confidence of the speaker did not predict listeners' recall accuracy or confidence in recall accuracy. Previous researchers have shown that confident speakers are more persuasive (e.g., Guyer et al., 2021); however, our speaker was not attempting to persuade; instead, she was attempting to inform. Being more persuasive might differ from being a better teacher. Future researchers may want to explore this idea further. An alternate explanation for speaker confidence not predicting recall is that our participants had low recall overall. Their inability to accurately respond to questions about the speech's content may reflect low motivation or lack of interest. These participants were young undergraduate students listening to a talk about research. They may not have seen the personal relevance of the topic. In support of this argument, the participants reported low levels of personal research experience. Some of the qualitative comments indicated that the audience felt the speech could have been more engaging (i.e., less boring).

In terms of recall accuracy and recall confidence, it is worth noting that listener characteristics were also not highly predictive. Listeners' empathy level, age, and race were unrelated to their recall. Compared to women, men had more accurate recall; however, men also reported more research experience than women. Thus, men's better recall may reflect their prior experience with the topic more so than their gender. (It is worth noting that perceptions of speaker attractiveness do not explain this gender difference.) Accuracy did not predict how listeners viewed the speaker, but listeners who had more confidence in their accuracy evaluated the speakers more positively. This finding might reflect affect infusion, where students who felt they personally learned something experienced positive emotions that they then generalized to the speaker.

We also hypothesized that greater cognitive empathy, the ability to understand others' feelings, would result in listeners' evaluating the low confidence speaker less harshly. This prediction was not supported. Instead, we found that listener's cognitive empathy did not influence their attribution of positive or negative traits to either speaker; the salient factor was the confidence of the speaker. Regardless of cognitive empathy level, the high confidence speaker was viewed more favorably. A deeper intellectual understanding of how the speaker was feeling was not strong enough to override the impressions created by the speech itself. The fact that listeners' cognitive empathy did not predict perceptions of the speakers matches previous researchers' argument that individual listener characteristics exert minimal influence (Brown et al., 2021b; Hancock et al., 2023; Marholz et al., 2018; Welch et al., 2021).

In contrast to cognitive empathy and failing to support our hypothesis, listeners who were higher in affective empathy, the ability to feel what others' feel, judged the low confidence speaker most negatively and the high confidence speaker least negatively. One explanation might be that personally relating to (i.e., taking on) the feeling of low confidence elicited unhappy feelings that led to harsher ratings, while personally relating to the feeling of high confidence elicited happy feelings that led to less harsh ratings. Another possibility is that affective empathy predicts emotion dysregulation (Thompson et al., 2022). The listeners with these more rapidly fluctuating emotions may have been more likely to use extreme negative ratings. In support of this argument, affective empathy impacted listeners' responses to negative attributes about the speakers but not their responses to positive attributes of the speakers. Again, it is noteworthy, and consistent with previous research, that listeners' characteristics exerted minimal influence on perceptions of the speakers.

To enhance the generalizability of our findings, future researchers could investigate whether these outcomes hold true for male versus female speakers or across a variety of speakers who differ in age, race, and cultural background. Additionally, a more nuanced understanding of variables that impact listeners' perceptions of speakers could be obtained by extending our protocol to examine the effect of nonverbal cues, including gestures, facial expressions, and body language. Future researchers might also examine how listeners' personal interest in the topic interacts with the studied variables. Jablon-Roberts and McCracken (2022) reported that undergraduate students have

overwhelmingly favorable impressions of guest speakers in the college classroom, so replicating this type of study in person rather than via video may render different results. Finally, future research could investigate the effects of speaker confidence in different contexts, such as work presentations, political speeches, and informal conversations, using a larger, more diverse sample.

In sum, we found that a confident female speaker was viewed more positively than an uncertain female speaker; however, confidence did not impact how much listeners recalled. Listeners' empathy did not predict their perceptions with one exception. Listeners higher in affective empathy evaluated the low confidence speaker most negatively and the high confidence speaker least negatively. Age, race, gender, and ethnicity did not emerge as highly predictive listener characteristics. These findings suggest that the confidence of female speakers may be even more influential than listener characteristics.

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Appendix

Script

I'm going to tell you about a research project that I recently did examining how college students perceived their high school teachers versus how they perceived their college teachers. The way that students perceive their teachers influences how well they get along with them. This is important because the more students get along with their teachers, the more they learn, the better grades they get, and the better behaved they are in class. Most of the research has focused on K-12 classrooms. My study added to what was previously known because I focused on college students.

I hypothesized that college students would report more positive interactions with their college teachers compared to their high school teachers. I also decided to look at academic entitlement. Students high in academic entitlement think that learning is the responsibility of the teacher, not their responsibility. They expect a lot of assistance and blame others when they fail.

I hypothesized that students who felt a lot of academic entitlement would have more negative interactions with all of their teachers.

I used survey methodology. First, I asked participants to think only about their high school teachers. I asked them to tell me what their teachers were like, how they interacted with them, and how they behaved in their classrooms. Then, I asked participants to think about their college teachers and answer the same questions.

When I analyzed the data, I found that overall, students viewed teachers positively; however, students attributed more positive traits to and were more willing to communicate with college teachers. College teachers were viewed as more professional and more sensitive to diversity. Students also said they lied and cheated more in high school than they did in college.

These results probably reflect students' different maturity levels and goals. It also matches my hypothesis!

My other prediction was that an attitude of entitlement would interfere with student-teacher relationships. This hypothesis was also correct! The more entitled students were, the less positive things they had to say about their high school and college teachers. They were also more likely to cheat and to complain about teachers to other teachers.

Last, I examined demographic characteristics. White students reported feeling more comfortable around high school teachers, while Black students were more likely to talk with their college teachers. This might have something to do with those earlier results I told you about, where college teachers were considered to be more sensitive to diversity. Men and women said the same thing about their high school teachers. However, in college, men were more likely to talk to their teachers, especially if they had something to complain about. I compared students who reported being heterosexual with those in other categories. Non-heterosexual students described a more informal interaction style with their high school and college teachers… for example, joking around and gossiping.

Thank you for allowing me to share my research findings. I hope you found it interesting and will maybe feel inspired to do research of your own…if you haven't already.